

SCI-NOI v0.1

SCI-NOI v0.1 r0.5 Shared Normative Core

Implementation-blind Stage B draft

Scientific owner: Grant Wilson

SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.5

Conditional owner-freeze candidate; not frozen. No implementation conformity, generator validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Document identity: SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.5

Scientific owner: Grant Wilson

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Status: implementation-blind proposed scientific-owner freeze candidate; scientifically freezeable conditional; not frozen.

The complete shared normative authority is the following six files in fixed order:

1. normative/NOTATION.md
2. normative/DEFINITIONS.md
3. normative/EQUATIONS.md
4. normative/ASSUMPTIONS.md
5. normative/REQUIREMENTS.md
6. normative/PREDICTIONS.md

They are byte-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.5 in NORMATIVE_MODULE_BINDING.json, binding-file SHA-256 aa59ecaaaa149e2990d07623563d90af76c7b3084ee37c497a06e17ebf0fe213.

The binding records the immutable r0.18 manifest and owner bytes, the accepted r0.3 owner supplement, and both later owner directives:

- Authority | SHA-256
- SCI-NOI_AUTHOR_PACKET_MANIFEST v0.1/r0.18 |
b6f8e7252e7f61f4506899cb3e8e26cf939887bb48464852713f8ce81ac77ca0
- SCI-NOI_OWNER_DECISIONS v0.1/r0.18 |
272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b
- SCI-NOI_OWNER_DECISION_SUPPLEMENT v0.1/r0.3 |
4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5
- SCI-NOI_OWNER_DIRECTIVE v0.1/r0.4 |
f155c1e70c1a8431bf4baf68f111960456ef24e9e77544837ff2efbecb86f423
- SCI-NOI_OWNER_DIRECTIVE v0.1/r0.5 |
4f04e25a32b44b729bd291d4919c9acb4bc197f79d94f7f213f4d9505539114d

The r0.5 modules preserve every r0.4 identifier, amend affected stable clauses, and append only REQ-048 through REQ-051 and PRED-023 through PRED-026. These close assignment-ignorable member selection; the separation of ideal target law, key/bit-selection law, and deterministic replay; conditional rather than unqualified UNC set/count equality; lossless rational-parent identity; and the distinct not_requested and explicitly_disabled lifecycles.

Any module byte change requires a new binding and regenerated documents. No rationale, ECS procedure, table, report, manifest, or PDF supersedes the modules. The normative-core PDF appends their exact text in order.

Authority, freeze, and claim boundary

The r0.18 profiles and Registry records remain immutable. Four revised @2 successor records are complete proposals only: owner approval and SCI-VAL source/Registry binding remain open, so changed-use evaluation remains unavailable. Deterministic replay establishes exact realized identity but not iid behavior, random-generator adequacy, or estimator uncertainty.

Unavailable exact numerical parents/routes, revised profiles, and complete STD response block their dependent products and claims. Implementation conformity and empirical validation are unassessed. These conditions do not prevent the implementation-independent scientific contract from being proposed for conditional owner freeze; this package does not itself freeze it.

No implementation conformity, random-generator validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, production suitability, or production authorization is established.

Appendix A. Exact Bound Normative Modules

SCI-NOI v0.1 r0.5 Normative Notation

Module identity: SCI-NOI_NORMATIVE_NOTATION v0.1/r0.5 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

- Symbol | Exact meaning
- i, d, h, b, p | Exact retained occurrence, observation-scoped detector/channel, exact network, member, and frozen MAP row identities.
- $Z_i^{\text{PTC}}, \gamma_i, G_{pi}, a_{pi}$ | PTC signal, exact MAP-facing coefficient, frozen projection, and positive contribution G_{pi} γ_i .
- C_p, Q_p | Exact contributing occurrences and unsigned normalization $\sum_i a_{pi}$.
- β_d, D_h^+ | Exact detector mass and $\{d: \beta_d > 0 \text{ and exact } h(d)=h\}$.
- ϵ_{bd}, τ_h | Active detector sign and exact no-default tolerance in $[0,1)$.
- $(n_x, q_x), \text{decode}$ | Canonical reduced arbitrary-precision rational for design quantity x and its lossless decoding; $q_x > 0$, gcd one, zero only $(0,1)$.
- A_h, A | Exact admitted assignment set for active stratum h , and $A = \text{product}_h A_h$ under canonical stratum concatenation.
- $K, F(K, \text{plan})$ | Bound opaque key and deterministic replay map from key/counters to the realized assignment sequence.
- R_b | Conjunction of binding completion and UNC-member-admission facts for member b ; fixed before assignment or invariant under it.
- $B_{\text{resolved}}, A_{\text{UNC}}$ | Actual resolved member-identity set and initial-UNC admitted set; equality is required only when initial UNC is realized.
- D_{ϵ_b} | Occurrence-diagonal expansion of the active assignment.
- M_b, m_{MAP} | NOI member and independently governed ordinary SCI-MAP signal.
- $\text{Vhat}_{\text{cond}}, \sigma_{\text{cond}}, \zeta_{\text{cond}}$ | Conditional marginal second moment, its square-root scale, and unit-one standardized MAP.
- $N_{\text{requested}}, N_{\text{resolved}}, N_{\text{completed}}, N_{\text{admitted}}$ | Actual requested, resolved, completed, and initial-UNC admitted counts; failed/unavailable states may have unequal values.
- $N_{\text{unique}_{\text{sign}}}, N_{\text{unique}_{\text{orbit}}}, r_{\text{sign}}, r_{\text{map}}$ | Distinct assignments, complement orbits, uncentered sign rank, and named projected rank.
- I_{attempt} | Execution-attempt identity, distinct from scientific identity.
- $I_{\text{request}}, I_{\text{effective}}$ | Exact request and effective-plan identities; absent for $\text{not}_{\text{requested}}$, both present for $\text{explicitly}_{\text{disabled}}$.
- $\text{initial}_{\text{UNC}}_{\text{realized}}$ | Typed realization state whose truth implies exact all-member set/count equality and estimator existence.

SCI-NOI v0.1 r0.5 Normative Definitions

Module identity: SCI-NOI_NORMATIVE_DEFINITIONS v0.1/r0.5 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

D-001 Operations and route

GEN, UNC, and STD are separate. The ordinary route inserts one active detector sign at the exact PTC-to-frozen-MAP boundary; NOI owns design and realization, MAP owns conforming frozen arithmetic.

D-002 Fixed method

Every consequential adjacent state is fixed; relearning or reselection creates another method and ensemble.

D-003 Population and scope

Base v0.1 has one ensemble per exact observation and TolTEC array. Complete, active, zero-mass, zero-total, singleton, and ambiguous states retain their approved typed dispositions.

D-004 Selected design

Independent symmetric candidates are first-accepted under every exact active-stratum condition, with positive cap, fail-closed exhaustion, replacement, valid duplicate draws, separate complement orbits, and no forced pair.

D-005 Disabled and enabled

not_requested has no requested GEN method, eligibility proposition, or assignment design. explicitly_disabled retains exact request/effective-plan identity plus disabling owner, policy, and cause, but no resolved assignment design. Both have $N_{\text{requested}}=0$ and no members or downstream product; they are not aliases. Enabled means a positive integer request and complete resolution; a smaller ensemble is never success.

D-006 Exact arithmetic

Design quantities use canonical reduced arbitrary-precision rational pairs and exact integer cross multiplication. Every a_{pi} rational is a lossless encoding of the exact frozen MAP contribution bound to its product/generation, G_{pi} , coefficient family/generation/payload, representation source, and digest. It governs NOI design only and never replaces MAP arithmetic.

D-007 Joint law

The scientific randomization measure, ideal candidate-bit/key-selection law, and deterministic replay map are distinct. Before key resolution the declared ideal model is $\epsilon_{\text{b}} \text{ iid} \sim \text{Uniform}(A)$, where A is the canonical product of stratum-admitted sets. After key binding, replay is deterministic and establishes identity, not statistical independence or generator adequacy.

D-008 Source-bearing target

The fixed parent may contain source and residual structure. The ensemble is not a source-free null, repeated physical noise, or calibrated tail law.

D-009 Initial UNC

The primary product is conditional_detector_sign_randomization_marginal_second_moment. Known zero target mean makes it equal to target-law marginal variance; it is not displayed as map/noise variance or total uncertainty.

D-010 Atomic UNC membership

Realized base UNC implies $A_UNC=B_resolved$ and equality of positive requested, resolved, completed, and admitted counts. Every resolved member completes, passes assignment-ignorable policy, shares the common domain, and enters the estimator. Failed or unavailable states retain factual possibly unequal counts and causes; no survivor normalization or estimator exists.

D-011 Effective information

Effective Monte Carlo information is estimator-, target-, domain-, dependence-, and calculation-specific. It is not inferred from counts, uniqueness, or ranks.

D-012 STD response

zeta_cond is nonlinear and data-dependent because numerator and scale descend from the same observed parent. A fixed-scale derivative is not the full procedure response.

D-013 Ownership and profiles

GEN owns completion; NOI owns use policy; VAL binds/evaluates. Immutable r0.18 profiles remain exact; changed r0.5 uses are unavailable until approved and bound successors exist.

D-014 Claim ceiling

No reciprocal ordinary product, significance, detection, conformity, random-generator validation, calibration, performance, readiness, freeze, or production claim is supplied.

D-015 Design-ignorable member selection

Every binding predicate in R_b is fixed before assignment or invariant under ϵ_b and M_b . Assignment-dependent diagnostics are advisory only. A required output-dependent gate defines a different selection-conditioned method; base v0.1 becomes unavailable rather than silently changing its target law.

D-016 Conditional scientific freeze

Exact unavailable numerical parents, routes, profile authority, full STD response, implementation conformity, and empirical validation block their dependent products and claims but do not prevent conditional freeze of a coherent implementation-independent contract.

SCI-NOI v0.1 r0.5 Normative Equations

Module identity: SCI-NOI_NORMATIVE_EQUATIONS v0.1/r0.5 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

SCI-NOI-EQ-001 $a_{pi}=G_{pi} \gamma_i$.

SCI-NOI-EQ-002 $Q_p=\sum_{i \text{ in } C_p} a_{pi}$.

SCI-NOI-EQ-003 $\beta_d=\sum_p \sum_{i \text{ in } C_p, d(i)=d} a_{pi}$; $D_h^+=\{d: \beta_d>0 \text{ and exact } h(d)=h\}$.

SCI-NOI-EQ-004 $\text{abs}(\sum_{d \text{ in } D_h^+} \epsilon_{bd} \beta_d) \leq \tau_h \sum_{d \text{ in } D_h^+} \beta_d$, $0 \leq \tau_h < 1$.

SCI-NOI-EQ-005 $M_b(p)=[\sum_{i \text{ in } C_p} a_{pi} \epsilon_{b,d(i)} Z_i^{\text{PTC}}]/Q_p$; the sign occurs exactly once only on Z_i^{PTC} , and every other MAP fact/gate is frozen.

SCI-NOI-EQ-006 $m_{\text{MAP}}(p)=[\sum_i a_{pi} Z_i^{\text{PTC}}]/Q_p$ only when independently governed SCI-MAP is scientifically realized; NOI cannot manufacture it by all-plus-one assignment.

SCI-NOI-EQ-007 $R_b^{\text{fixed}}=A_{\text{MAP},Pi} D_{\epsilon_b} H_{\text{PTC}}^{\text{fixed}}$; required response companions have identical membership/sign occurrence.

SCI-NOI-EQ-008 $E_D[\epsilon_{bd}]=0$ for active d ; $E_D[M_b(p)|\text{fixed parent/operator}]=0$ on exact available rows.

SCI-NOI-EQ-009 $\text{initial_UNC_realized}$ implies $A_{\text{UNC}}=B_{\text{resolved}}$ and $N_{\text{admitted}}=N_{\text{completed}}=N_{\text{resolved}}=N_{\text{requested}}>0$. No equality is asserted for a failed or unavailable UNC state.

SCI-NOI-EQ-010 When $\text{initial_UNC_realized}$, $V_{\text{hat_cond}}(p)=(1/N_{\text{resolved}}) \sum_{b \text{ in } B_{\text{resolved}}} M_b(p)^2$ on the exact common all-member domain. Otherwise no estimator or changed divisor exists.

SCI-NOI-EQ-011 For canonical stratum order, $A=\text{product}_{h \text{ active}} A_h$. Before key resolution, conditional on complete assignment-design resolution, the declared ideal model is $\epsilon_1, \dots, \epsilon_N \text{ iid } \sim \text{Uniform}(A)$. After binding exact key K , the realized sequence is deterministically $\epsilon_1, \dots, \epsilon_N=F(K, \text{plan})$.

SCI-NOI-EQ-012 For rational left numerator $L_h=n_L/q_L$, total $T_h=n_T/q_T$, and $\tau_h=n_{\tau}/q_{\tau}$, admission is exact integer comparison $n_L q_T q_{\tau} \leq n_{\tau} n_T q_L$ after nonnegative canonical reduction.

SCI-NOI-EQ-013 $\sigma_{\text{cond}}=\sqrt{V_{\text{hat_cond}}}$ and $\zeta_{\text{cond}}=m_{\text{MAP}}/\sigma_{\text{cond}}$ on the exact finite-positive compatible support.

SCI-NOI-EQ-014 $\delta \zeta_{\text{fixed_scale}}=\text{diag}(1/\sigma_{\text{cond}}) \delta m_{\text{MAP}}$, only with scale explicitly held fixed.

SCI-NOI-EQ-015 $\delta \zeta=\delta m_{\text{MAP}}/\sigma_{\text{cond}} - m_{\text{MAP}} \delta V_{\text{hat_cond}}/(2 \sigma_{\text{cond}}^3)$; the second term requires separately authorized complete-procedure response.

SCI-NOI-EQ-016 $\text{decode}(n_{pi}/q_{pi})=a_{pi}^{\text{MAP}}$ exactly, with the rational bound to the exact frozen MAP product/generation, G_{pi} , γ_i family/generation/payload, representation source, and source digest.

SCI-NOI-EQ-017 For every e in A , $P(R_b=1 | \epsilon_b=e, \text{fixed parent and plan})=P(R_b=1 | \text{fixed parent and plan})$; deterministic predicates may satisfy this by exact invariance.

SCI-NOI v0.1 r0.5 Normative Assumptions

Module identity: SCI-NOI_NORMATIVE_ASSUMPTIONS v0.1/r0.5 Scientific owner: Grant Wilson. Status: proposed final; not frozen.

SCI-NOI-ASM-001 Authority is the exact r0.18 packet, r0.3 owner supplement, r0.4 and r0.5 owner directives, and this byte-bound module set; process records add no science. SCI-NOI-ASM-002 Numerical GEN remains unavailable until exact PTC coefficient/QC, canonical rational design values, numerical coverage_cut, MAP admission, response, profiles, and every parent/plan fact exist. SCI-NOI-ASM-003 The real signal and every member share identical occurrences, operator, denominator, support, response, and gates. SCI-NOI-ASM-004 Independent symmetric candidate bits belong to the declared ideal randomization model. Exact content-bound disjoint counters and keying supply deterministic replay identity; replay success alone supplies no independence or generator-quality validation. SCI-NOI-ASM-005 First acceptance conditional on success is uniform on A; the finite cap changes resolution probability only. SCI-NOI-ASM-006 Candidate rejection is construction, not a member or failure. Binding completion and member-use predicates are fixed before assignment or assignment-invariant; assignment-dependent failure makes the base method unavailable and cannot select a favorable survivor ensemble. SCI-NOI-ASM-007 The fixed parent may contain source and residuals; no physical-noise or significance model is supplied. SCI-NOI-ASM-008 r0.18 profiles and Registry objects remain immutable. Proposed r0.5 successors are unevaluable pending owner approval and binding. SCI-NOI-ASM-009 Adjacent parents remain immutable; transforms and relearning require external authority and new generations. SCI-NOI-ASM-010 Full STD response is unavailable until the complete conditional-second-moment response is separately authorized. SCI-NOI-ASM-011 The ideal unconditioned Uniform(A) target is retained only under EQ-017. A scientifically required assignment-dependent gate defines a different conditioned law, method, estimator, weights, uncertainty, and claim. SCI-NOI-ASM-012 Numerical estimator uncertainty remains unavailable until an exact method supplies it; neither deterministic replay nor a single realized assignment set establishes that uncertainty.

SCI-NOI v0.1 r0.5 Shared Normative Module: Requirements

Module identity: SCI-NOI_NORMATIVE_REQUIREMENTS v0.1/r0.5

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by
SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.5; not frozen.

shall is required; shall not is prohibited; not_applicable and unavailable are typed scientific states, not numerical sentinels.

SCI-NOI-REQ-001 GEN, UNC, and STD shall remain separate operations. None shall automatically realize, authorize, or mutate another.

SCI-NOI-REQ-002 The ordinary route shall be NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1; signed outputs shall be NOI realizations, not ordinary MAP science products.

SCI-NOI-REQ-003 Every GEN method shall classify consequential adjacent state as fixed, rerun/relearned, not applicable, or unavailable. Different graphs shall not share an ensemble.

SCI-NOI-REQ-004 The coherence unit shall be one stable detector/channel in one exact observation. Its sign shall be constant across all admitted occurrences and ordered by canonical scientific identity.

SCI-NOI-REQ-005 Network balance shall use frozen positive coefficient mass β_d . No network, array, or observation shall balance another, and β_d shall not be promoted to another scientific role.

SCI-NOI-REQ-006 Base v0.1 shall produce one independent observation-level ensemble per exact observation and TolTEC array. It shall define no cross- observation randomized ensemble, randomized coadd, or multi-observation GEN route. Any future route requires the separate contract listed in the r0.3 owner supplement.

SCI-NOI-REQ-007 Every ordinary member shall satisfy SCI-NOI-EQ-005. The sign shall act only on Z_i^{PTC} , exactly once; all other MAP facts and gates shall remain frozen.

SCI-NOI-REQ-008 Inline and materialized application shall be equivalent only for identical equation, occurrence membership, assignment, frozen operator, and output domain. All-+1 arithmetic shall not manufacture an SCI-MAP product.

SCI-NOI-REQ-009 SCI-NOI-EQ-007 shall govern the fixed signal/response relation. Required response companions shall use identical membership and the modifier exactly once; missing response shall be unavailable without hidden signal subsetting.

SCI-NOI-REQ-010 The complete population, D_h^+ , zero-mass detectors, zero-total strata, singleton active strata, and ambiguous positive-mass membership shall remain separately typed. No zero-denominator ratio shall be evaluated.

SCI-NOI-REQ-011 Zero-mass detectors shall be excluded from the active sign matrix but retained as typed facts. Zero-total strata shall be not_applicable. Ambiguous positive-mass membership and every singleton active stratum shall make the selected balance design unavailable; neither shall be silently exempted or cross-balanced.

SCI-NOI-REQ-012 The selected ODQ-102D design shall use exact τ_h in $[0,1)$ without default, independent symmetric Rademacher candidates, first-accepted bounded rejection, a positive exact preregistered cap, fail- closed exhaustion, with-replacement draws, equal weight $1/N_{\text{resolved}}$, scheduling-independent content-bound generator/keying, and no tolerance relaxation or best-failed fallback.

SCI-NOI-REQ-013 Every design shall bind an exact target probability law or deterministic finite measure; stable population/order; assignment alphabet; probabilities/weights; equality, duplicate, complement, and equivalence relations; requested/resolved/completed/admitted/unique counts; applicable ranks/null spaces; deterministic reconstruction identity; lifecycle, generation, completion, and failure. Every applicable field shall be preregistered before candidate or member output inspection. Unused fields shall be not_applicable.

SCI-NOI-REQ-014 RNG identity and version, opaque seed/key bytes, namespace, canonical serialization, stable ordering, observation/array/stratum/member/ attempt counters, domain separation, tolerance, retry cap, and rejected- candidate count shall be required only for a selected stochastic rejection family. For the selected family they shall be exact and plan-bound. Conditional on success, finite first-accepted rejection shall be uniform over the admitted vectors; the cap shall change resolution probability only.

SCI-NOI-REQ-015 Under every accepted design, exact duplicate equality and exact complement-orbit relation shall be represented separately. Under the owner-selected independent with-replacement rejection family, complement- related assignments shall remain distinct and no complement pair shall be forced.

SCI-NOI-REQ-016 Exact duplicates shall remain distinct draws. Products shall report all six N_* counts, r_{sign} , r_{map} , and use-specific effective information separately; none shall substitute for another.

SCI-NOI-REQ-017 The source-bearing conditional randomization ensemble shall carry D-008's objective and exclusions. Zero target-law mean shall not imply source removal from members, moments, or tails.

SCI-NOI-REQ-018 Every admitted assignment shall complete through the exact frozen operator. Any assignment-dependent execution or numerical failure shall make the exact method unavailable. The design shall not be regenerated under the same method identity until a favorable all-pass ensemble appears; survivors and partial accumulators are diagnostic only.

SCI-NOI-REQ-019 An idempotent retry may retain scientific identity only for identical deterministic work with a new I_{attempt} . Changed scientific state shall be a replacement with new identity.

SCI-NOI-REQ-020 Each persistence plan shall select persisted, compact-regeneration, streaming-sufficient-statistic, or a separately identified composite and shall declare reconstruction, regeneration, sufficiency, unavailable later questions, lifecycle, completion, and failure.

SCI-NOI-REQ-021 An external deterministic transform shall require exact owner authority and parity for every operator fact; NOI shall not choose, tune, relocate, simplify, substitute, or silently omit it.

SCI-NOI-REQ-022 Wiener or FRUIT learning, selection, continuation, update, or per-member replay shall create every applicable new immutable owner, product, GEN, and UNC generation. Base numerical transformed routes remain unavailable.

SCI-NOI-REQ-023 Every UNC method shall bind exact target law, GEN membership, center, estimator, weights, dependence/missingness, domain, WCS/support/ response/unit, representation, ranks/null spaces, regularization, omissions, estimator uncertainty, lifecycle, claim, and named use.

SCI-NOI-REQ-024 Realized initial UNC shall consume only the exact resolved ensemble and shall imply $A_{\text{UNC}} = B_{\text{resolved}}$ and $N_{\text{admitted}} = N_{\text{completed}} = N_{\text{resolved}} = N_{\text{requested}} > 0$. Every resolved member shall complete, pass a member policy fixed before assignment or proven assignment-invariant, belong to the common domain, and enter the estimator. In failed or unavailable states, actual counts and causes shall remain factual and may be unequal; no estimator, survivor subset, renormalization, or changed divisor shall exist.

SCI-NOI-REQ-025 Initial UNC shall compute SCI-NOI-EQ-010 on the exact common domain with divisor N_{resolved} . It shall use the known zero target center and shall not subtract the finite ensemble mean, apply B-1, or replace the full resolved ensemble by a successful subset.

SCI-NOI-REQ-026 The primary product name shall be `conditional_detector_sign_randomization_marginal_second_moment`, with squared signal units. The equality to target-law marginal variance shall be stated separately; prohibited shortened names and physical-noise/total-uncertainty/ covariance/precision/significance claims shall not be used.

SCI-NOI-REQ-027 Numerical estimator uncertainty shall be explicit or unavailable. Counts, uniqueness, ranks, and effective information shall remain separate from that uncertainty.

SCI-NOI-REQ-028 Retained ensemble, marginal moment, projection, structured/full covariance, reciprocal, precision, weight, and unavailable representations shall remain separate roles. Missing covariance is unknown, not zero or independence.

SCI-NOI-REQ-029 No reciprocal shall enter the ordinary base bundle without a new exact owner disposition. Every reciprocal, inverse variance, precision, and consumer weight shall remain unavailable; STD shall consume `sqrt(Vhat_cond)` directly. Any future reciprocal shall use the exact name `reciprocal_conditional_randomization_second_moment`, finite-positive domain, separate identity/profile/use, and no prohibited promotion.

SCI-NOI-REQ-030 Initial STD shall use independently realized `m_MAP`, `sigma_cond`, and `zeta_cond` on the exact compatible finite-positive intersection. Missing MAP coefficient, numerical coverage_cut, MAP admission, or other required parent shall make `m_MAP` and STD unavailable. JINC remains separate and unavailable.

SCI-NOI-REQ-031 `zeta_cond` shall have unit 1 and the sole claim: "MAP signal standardized by the stated conditional detector-sign-randomization second-moment scale." No significance, probability, detection, false-alarm, completeness, purity, or catalog claim shall follow.

SCI-NOI-REQ-032 Admission shall retain separately named request, applicability, eligibility, and realization fields. Only the exact positive conjunction shall project to the exact named action.

SCI-NOI-REQ-033 Generation-input admission shall admit only one exact candidate occurrence for the selected GEN method; later operations remain separate.

SCI-NOI-REQ-034 UNC-member admission shall consume but not redefine GEN completion, failure, duplicate/equivalence, support, imprint, QC, persistence, lifecycle, cause, and provenance facts. Every binding predicate shall be fixed before assignment or proven invariant under `epsilon_b` and `M_b`; output amplitude, sign, imprint, morphology, tail value, or assignment-dependent QC shall be advisory and shall not exclude a base-method member.

SCI-NOI-REQ-035 UNC-ensemble admission shall admit only one exact complete all-members-successful, design-ignorable ensemble for one estimator/domain and shall not realize the estimator or authorize another use. A failed design shall not be repeatedly regenerated under the same scientific identity until an all-pass selection occurs.

SCI-NOI-REQ-036 STD admission shall permit only the exact initial numerator/ scale join and claim. The proposed r0.5 successor profile has no authority by name alone; evaluation shall remain unavailable until exact owner approval and Registry/source binding.

SCI-NOI-REQ-037 Missing, conflicting, incomplete, unsupported, invalid, or failed required facts shall yield typed states. No undocumented sentinel, filename, shape, approximate WCS, or hidden default shall establish success.

SCI-NOI-REQ-038 PTC, MAP, JINC, transform, Wiener, FRUIT, and prior NOI parents shall remain immutable; later results shall be new versioned companions with exact ancestry and lifecycle.

SCI-NOI-REQ-039 No assignment, uncertainty, reciprocal, precision, scale, or weight shall become validity, support, exposure, or a PTC/MAP coefficient without separately approved successor authority.

SCI-NOI-REQ-040 Existing r0.18 profile and Registry objects shall remain immutable. The four complete proposed r0.5 successor records shall remain proposals, not executable authority. Affected evaluation shall remain unavailable until exact successor approval and Registry/source binding.

SCI-NOI-REQ-041 The rationale and ECS shall bind the exact six shared modules and shall not independently author equations, requirements, or predictions. Any module byte change requires a new binding and document generation.

SCI-NOI-REQ-042 This draft shall establish no implementation conformity, numerical or random-generator validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization.

SCI-NOI-REQ-043 Both `not_requested` and `explicitly_disabled` shall have `N_requested=0`, no resolved assignment design, members, successful empty ensemble, UNC, or STD. They shall retain the distinct identities in D-005. Enabled GEN shall require an exact positive integer `N_requested`; success shall require `N_resolved=N_requested>0`. Any member-assignment resolution failure within the preregistered cap shall fail the complete design, never realize a smaller successful ensemble. A rejected candidate is neither a member nor a member failure.

SCI-NOI-REQ-044 Every balance input `a_pi`, threshold `tau_h`, imbalance numerator and denominator, and admission comparison shall use canonical reduced arbitrary-precision rational pairs (n,q) , with $q > 0$, $\gcd(|n|,q)=1$, zero as $(0,1)$, positive `a_pi` having $n > 0$, and $0 \leq \tau_h < 1$. Serialization shall be canonical ASCII n/q ; accumulation shall be exact in stable identity order; admission shall use exact cross multiplication. The exact `a_pi` rational shall satisfy EQ-016. A missing lossless exact rational shall make design resolution unavailable. Floating conversion shall not decide admission.

SCI-NOI-REQ-045 Before key resolution and conditional on complete assignment- design resolution, the declared ideal model shall satisfy $\epsilon_{b,iid} \sim \text{Uniform}(A)$ with $A = \text{product}_h A_h$ under canonical stratum concatenation. Candidate signs shall be ideal independent symmetric Rademacher values before conditioning, with disjoint stratum/member domains, replacement, valid distinct duplicates, and unforced distinct complements. After exact key binding, assignments shall be deterministic $F(K, \text{plan})$. Replay shall establish identity, not empirical independence, generator adequacy, or estimator uncertainty.

SCI-NOI-REQ-046 `N_resolved` shall denote conditional Monte Carlo draws, not exposure, observations, or automatic effective sample size. Effective Monte Carlo information shall not be inferred from member count, unique count, duplicate count, complement-orbit count, or sign/map rank. Every effective- information product shall name its estimator, target, domain, dependence model, and calculation, or be unavailable.

SCI-NOI-REQ-047 Initial STD shall use $\sigma_{\text{cond}} = \sqrt{V_{\text{hat_cond}}}$ and $\zeta_{\text{cond}} = m_{\text{MAP}}/\sigma_{\text{cond}}$. The fixed-scale derivative $\delta \zeta_{\text{fixed_scale}} = \text{diag}(1/\sigma_{\text{cond}}) \delta m_{\text{MAP}}$ shall be labeled as a held-scale partial response, never the full response. The full response shall be $\delta \zeta_{\text{cond}} = \delta m_{\text{MAP}}/\sigma_{\text{cond}} - m_{\text{MAP}} \delta V_{\text{hat_cond}} / (2 \sigma_{\text{cond}}^3)$ and shall remain unavailable until the second term has separate exact authority. Every response product shall carry dependence, response, support, and cause semantics; dividing a MAP response by `sigma_cond` shall not establish the full STD response.

SCI-NOI-REQ-048 Base-v0.1 member selection shall satisfy EQ-017: every binding completion and UNC-member-admission predicate in `R_b` shall be fixed before assignment or proven invariant under assignment and member output. Advisory output diagnostics shall not alter membership. If an assignment- dependent gate is scientifically required, it shall define a different method with an explicitly selection-conditioned target law, estimator, normalized weights, uncertainty, and claim; the unconditioned $\text{Uniform}(A)$ claim shall not be retained.

SCI-NOI-REQ-049 The scientific randomization measure, the ideal key/candidate- bit selection law, and deterministic replay map shall remain separate exact objects. Base v0.1 adopts the ideal iid-uniform model while placing random- generator and randomness adequacy in a separately named validation layer. A successful deterministic replay or one realized assignment set shall not count as implementation conformity, independence evidence, or generator validation.

SCI-NOI-REQ-050 Each rational design contribution shall be a lossless canonical lift satisfying $\text{decode}(n_{\pi}/q_{\pi}) = a_{\pi}^{\text{MAP}}$ and shall bind the exact frozen MAP product/generation, G_{π} , γ_i family/generation/payload, representation source, and digest. It shall not be rounded from display text, independently quantized, re-estimated, converted through undocumented precision, or taken from another generation. It governs NOI balance identity only and shall not replace independently governed MAP numerical arithmetic.

SCI-NOI-REQ-051 `not_requested` and `explicitly_disabled` shall not be aliases. The former shall have request axis `not_requested` and no GEN method, eligibility proposition, request identity, effective-plan identity, or assignment design. The latter shall retain an exact request/effective-plan identity plus effective state disabled, disabling owner, policy, and cause, while having no resolved assignment design or downstream product.

SCI-NOI v0.1 r0.5 Shared Normative Module: Predictions

Module identity: SCI-NOI_NORMATIVE_PREDICTIONS v0.1/r0.5

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.5; not frozen. These are analytic expectations, not implementation-conformity or empirical-validation reports.

SCI-NOI-PRED-001 A globally complemented active assignment has equal target probability under the approved complement-symmetric law.

SCI-NOI-PRED-002 Every active detector has target-law sign mean zero; the fixed linear member has target-law mean zero on each exact available row.

SCI-NOI-PRED-003 Complement symmetry does not remove source or structured residual contributions from individual members, second moments, or tails.

SCI-NOI-PRED-004 Complementing every active sign negates M_b while leaving the unsigned Q_p unchanged.

SCI-NOI-PRED-005 Complement-related members have identical pointwise squares and outer products while remaining distinct draws; exact duplicate and orbit relations remain separately represented.

SCI-NOI-PRED-006 $V_{\text{hat_cond}}$ is nonnegative wherever available and is unavailable outside the exact common all-member domain.

SCI-NOI-PRED-007 Source-bearing fixed structure can appear in $V_{\text{hat_cond}}$; zero target-law mean does not prevent it.

SCI-NOI-PRED-008 Any failed admitted member makes initial UNC unavailable; completed survivors do not realize a partial estimator.

SCI-NOI-PRED-009 More members may improve estimation of the declared randomization target but cannot create exposure or reduce the immutable parent's realized noise as $1/\sqrt{N}$.

SCI-NOI-PRED-010 A fixed owner-defined linear transform propagates a conditional moment under its exact operator; a learned per-member transform does not satisfy that identity by implication.

SCI-NOI-PRED-011 A positive profile decision neither realizes the next operation nor changes producer-owned facts.

SCI-NOI-PRED-012 Missing required response makes response-dependent output unavailable; signal membership does not silently shrink.

SCI-NOI-PRED-013 An idempotent retry may change attempt/evidence identity without changing scientific identity; a changed assignment or scientific state cannot.

SCI-NOI-PRED-014 Conditional on successful resolution, first-accepted finite rejection from independent symmetric candidates is uniform over admitted sign vectors: every admitted vector has the same base probability, and the common geometric probability of earlier rejection cancels in the conditional ratio. The cap changes success probability, not the accepted-vector law.

SCI-NOI-PRED-015 A zero-total-mass stratum is not_applicable to balancing and produces no imbalance ratio.

SCI-NOI-PRED-016 A singleton active stratum makes the selected balance design unavailable for every admissible τ_h ; it is neither exempted nor cross-balanced.

SCI-NOI-PRED-017 ζ_{cond} has unit 1 on its available domain; dimensionlessness establishes no pivotal, significance, probability, detection, false-alarm, completeness, purity, or catalog interpretation.

SCI-NOI-PRED-018 Either exact zero-cardinality lifecycle state produces no resolved assignment design or downstream product; an enabled successful request has exactly $N_{\text{resolved}} = N_{\text{requested}} > 0$; and failure to resolve any requested member within its cap produces a failed complete design, never a smaller successful ensemble.

SCI-NOI-PRED-019 If every resolved member completes but one member is ineligible for the exact common-domain UNC estimator, the complete initial UNC is unavailable; the remaining members are not renormalized into an estimator.

SCI-NOI-PRED-020 Exact reduced-rational cross multiplication admits equality at the balance boundary and rejects a value strictly above it, even when a finite floating representation rounds those two cases to the same comparison result.

SCI-NOI-PRED-021 Under the declared ideal model and conditional on complete resolution, members are independent uniform draws from the admitted assignment set with replacement. Therefore an exact duplicate is a valid distinct draw, and a complement-related assignment is a distinct possible draw rather than a forced partner.

SCI-NOI-PRED-022 Holding $V_{\text{hat_cond}}$ fixed removes the scale-response term and gives the stated fixed-scale partial derivative. When $V_{\text{hat_cond}}$ responds to the perturbation, the full derivative contains the additional $-m_{\text{MAP}} \Delta V_{\text{hat_cond}} / (2 \sigma_{\text{cond}}^3)$ term; without separately authorized $\Delta V_{\text{hat_cond}}$, the full STD response is unavailable.

SCI-NOI-PRED-023 If all binding R_b predicates satisfy EQ-017, conditioning on completion/admission leaves the ideal Uniform(A) assignment law unchanged. An output-dependent exclusion with unequal pass probabilities over A changes that law; base UNC becomes unavailable rather than estimating from selected survivors.

SCI-NOI-PRED-024 Replaying the same exact key, counters, serialization, and plan yields the same realized assignment sequence, but that equality alone does not establish iid behavior, random-generator adequacy, or estimator uncertainty.

SCI-NOI-PRED-025 A fixture whose rounded display decimal lies on the balance boundary while its lossless frozen-parent rational lies strictly above it is rejected by the exact design; using the display decimal would incorrectly admit the assignment.

SCI-NOI-PRED-026 `not_requested` and `explicitly_disabled` both have $N_{\text{requested}}=0$ and no downstream products, yet only the latter has request and effective-plan identities plus disabling owner, policy, and cause; their lifecycle records therefore differ falsifiably despite equal cardinality.